

Project Name: Teacher Insight Engine (TIE)

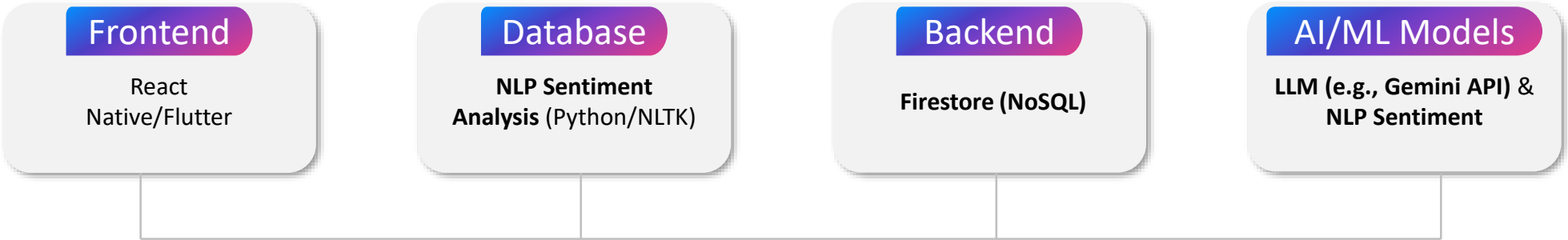
Tagline: Ethical AI for Teacher Growth, Beyond Test Scores.

Problem Statement No.: Problem Statement 4:
Teacher Performance Evaluation Gap

Team Members:

Vishwa Prakash	→	Team Leader & Researcher
Priyam Tiwari	→	Data Pipeline Leader
Shreya Singh	→	AI Logic Leader
Vaishali	→	Frontend UI Leader

Tech Stack Overview



Problem Statement & Technical Challenges

Core Problem:

Current evaluation relies heavily on **Student Test Scores** and **Subjective Observation**, which is **unfair, stressful, and incomplete**. It measures the *result*, not the *teaching technique*.

Success Metrics:

- **Evaluation Equity:** 30% reduction in evaluation bias.
- **Teacher Adoption:** Tool must be seen as a Coach, not a Monitor.

Technical Constraints:

- **Bias Mitigation:** AI must actively counter cultural and algorithmic bias.
- **Low-Resource Compatibility:** Solution must be efficient (avoiding heavy video/sensor data).
- **Privacy:** All data must be anonymized.

Users:

- Rural Teachers
- School Administrators
- Educational NGOs

Solution Overview

Elevator Pitch:

TIE (Teacher Insight Engine) is an **ethical AI coach** that analyzes **in-classroom content clarity and structure** (via audio) and fuses it with **sanitized student feedback** to generate actionable, growth-focused scores.

Key Differentiator:

1. **Ethical-First Design:** Mandatory Bias Mitigation Framework.
2. **Pedagogy Focus:** Ignores test scores; rewards teaching effort.
3. **Actionable Feedback:** Provides time-saving, specific advice.

Expected Impact:

Converts evaluation from a punitive process to a **growth cycle**, leading to **50% faster teacher skill development**.



Solution Architecture Overview

Data Flow

Data flows from the **Teacher's Phone** (Audio) → Transcription → **AI Lead** (Scoring Logic) → **UI Lead** (Dashboard). **Feedback** data is merged into the flow at the AI Engine step.

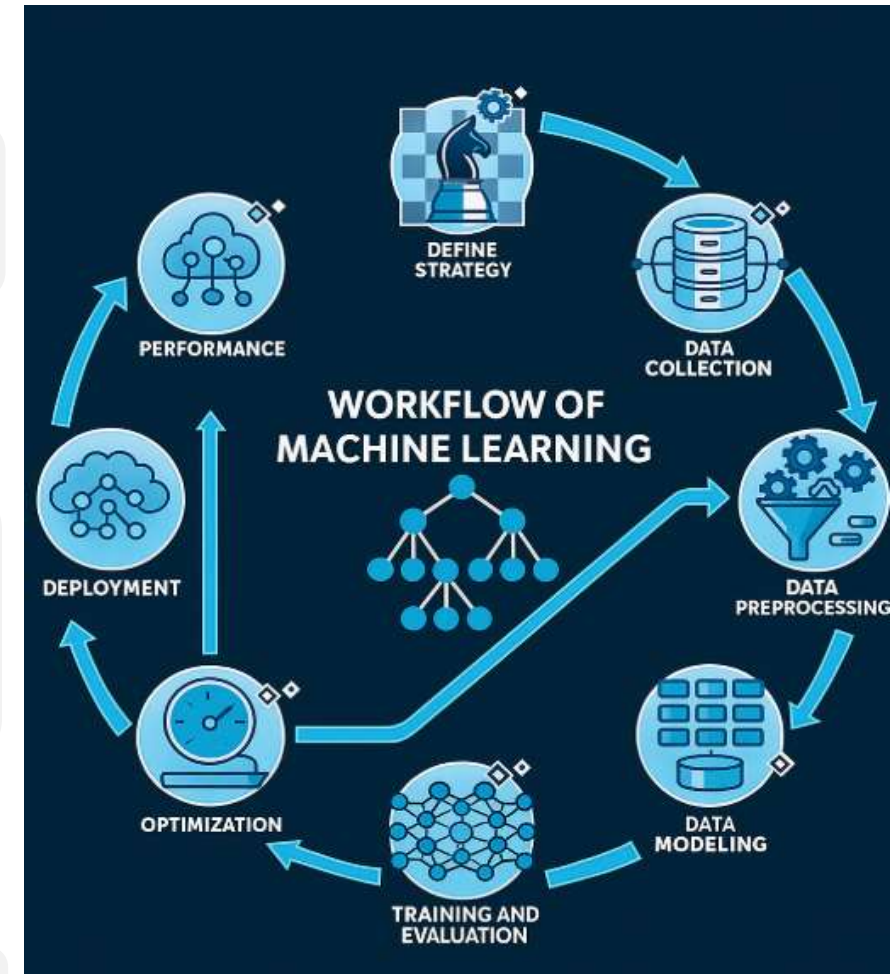
Cloud vs. Local Processing

Cloud (Serverless/LLM API): Used for **heavy processing** (Transcription, LLM Scoring) to ensure **scalability** and **speed**.

Local (On-Device Storage): Used to store basic historical data and run the feedback form, ensuring **offline-first access** to past reports

Scalability Approach

The entire backend runs on a **Serverless Architecture** (e.g., Firebase Cloud Functions).



Technology Stack



Frontend

React Native / Flutter

Allows development for both iOS and Android from a single codebase (faster MVP) and provides better **low-RAM** device performance than a web app.



Backend

Python / FastAPI

Python is the industry standard for AI/ML integration. FastAPI is chosen because it is **lightweight, fast, and has a low memory footprint**, which is crucial for low-bandwidth environments.



Database

Firestore (NoSQL)

Chosen for its **flexibility** (allowing us to easily change data structure as we evolve the product in an Agile manner) and excellent integration with our serverless backend.



AI/ML Core

LLM (e.g., Gemini API) & NLP Sentiment

LLM is essential for **complex, prompt-based scoring** (Clarity, Coverage). **NLP Sentiment** is mandatory for the **Bias Mitigation Framework** and filtering emotional input.

AI/ML Pipeline Architecture

Pipeline 1

Teaching Content Analysis (TCA) Flow:

1. **Ingestion:** Teacher Audio Recording.
2. **Pre-processing:** Audio-to-Text Transcription.
3. **Core Analysis (LLM Prompting):** Transcript is sent to the LLM API to generate **Clarity Score (1-10)** and **Concept Coverage (%)**.

Pipeline 2

Feedback Fusion & Fairness Check:

1. **Input:** Anonymous Student Feedback (Text).
2. **Bias Mitigation:** NLP Sentiment Analysis to **filter extreme emotion/bias**.
3. **Fusion:** TCA Scores aur Sanitized Feedback ko combine to generate **Final Evaluation Score**.

Key Advantage: Simple, Text-Based Processing ensures high performance on low-resource devices.

Data Architecture

Data Models

- Lessons (Audio File URL, Anonymized Transcript)
- Reports (Clarity Score, Coverage Score, Actionable Tips)
- Feedback (Raw Feedback Text, Sentiment Score)

Privacy Guardrail:

All sensitive data is **anonymized** and stored securely.

Storage Strategy

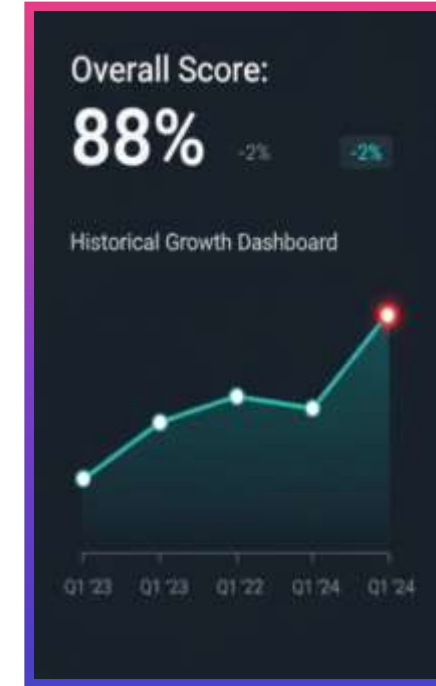
- **Cloud Database (Firestore):** Scalable user, lesson, and for report data.
- **Local Storage/SQLite:** Low-latency and offline dashboard access, storing basic performance history on the teacher's device.

UI/UX Design Approach

"Actionable, Transparent, and Low-Cognitive Load." (Easy and Fast).

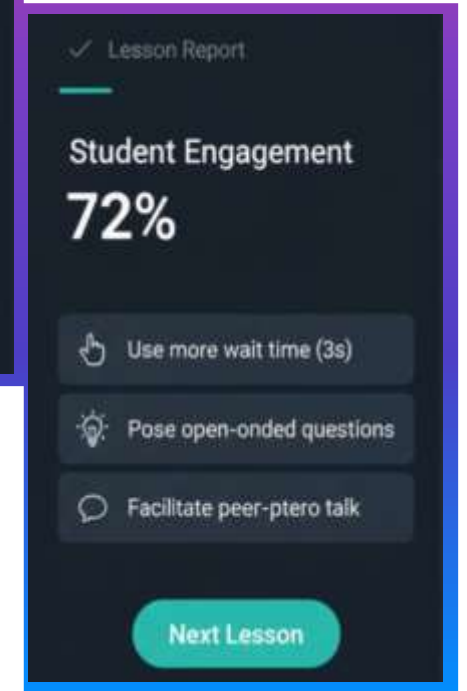
USER FLOW:

Teacher logs in → Uploads/Records Audio → Views **Lesson Report** → Gets **Actionable Tips**.



Historical Growth Chart

Every lesson has specific improvement tips





Performance Key Strategy

01

Text-Heavy, Media-Light: Our AI only analyzes text (transcript), **no heavy video processing.**

02

Lazy Loading: Data will only load on the dashboard when necessary.

03

Background Sync: Audio upload and score calculation will happen in the background so the user's app doesn't slow down.

Team Members & Roles



**VISHWA
PRAKASH**

**Team Leader
&
Researcher**

Overall project
direction,
strategy, and
research integrity.



**PRIYAM
TIWARI**

**Data Pipeline
Leader**

Managing the flow,
cleaning, and
preparation of data
for the AI/ML
system.



**SHREYA
SINGH**

**AI Logic
Leader**

Designing and
implementing the core
Artificial Intelligence
and machine learning
logic.



VAISHALI

**Frontend UI
Leader**

Overseeing the
development of the
user interface,
accessibility, and overall
user experience.

FUTURE SCOPE

1. Personalized PD Recommendations: AI suggests specific training modules or resources based on the teacher's TIE report (e.g., recommending the video we discussed earlier).
2. Parent/Admin Report Generation: Automatically generating simple, visual reports for school administrators to track overall school growth trends.

LONG TERM CHALLENGES

1. Integrating Multi-Speaker Diarization: Developing a custom solution for identifying teacher vs. student voices without relying on high-cost cloud APIs.
2. Cross-Cultural Pedagogy: Adapting the scoring logic to different regional teaching styles (e.g., teaching in UP vs. teaching in Kerala).

Conclusion

Key Achievements & Impact

- **Ethical AI Coaching:** Transforms evaluation with AI-driven, bias-mitigated feedback focusing on instructional quality.
- **Accessible & Scalable:** Low-resource compatible architecture ensures reach in rural and low-connectivity environments.
- **Empowering Growth:** Provides *actionable insights* that significantly accelerate teacher skill development.

Quantifiable Promises

- **30% Reduction in Evaluation Bias**
- **50% Acceleration in Teacher Skill Development**

Future Vision

- **Integrated into Medha AI:** A foundational tool for a robust educational ecosystem.
- **Future Enhancements:** Includes Speaker Diarization and personalized learning pathways.

THANK
YOU